

Project: MediaHub

Digital Content & Subscription Management Platform

Domain: Media, Entertainment & Publishing

1. Introduction

MediaHub is a web-based digital content and subscription management platform designed for media companies, OTT operators, publishing houses, and content studios to manage content catalogs, creator onboarding, subscription plans, audience engagement, content licensing, royalty calculations, and revenue analytics. The system enables content creators to upload and manage their catalog, subscribers to browse and manage plans, editorial teams to curate and schedule content, rights managers to track licensing and royalties, and administrators to configure subscription products, content policies, and reporting parameters.

It uses a REST API-based backend architecture with Angular/React frontend and is compatible with Java (Spring Boot) and .NET (ASP.NET Core) platforms.

Actors / Users

- **Subscriber / Viewer:** Browses content catalog, manages subscription plan, and tracks viewing history
- **Content Creator / Publisher:** Uploads content, manages catalog, and monitors performance metrics
- **Editorial Manager:** Reviews, curates, and schedules content for publication and featured placement
- **Rights & Licensing Manager:** Manages content licenses, territory rights, and royalty agreements
- **Revenue Analyst:** Tracks subscription revenue, content monetisation, and royalty payouts
- **Media Admin:** Configures subscription plans, content categories, licensing templates, and user roles

2. Module Overview

- **2.1 Identity & Access Management:** Authentication, RBAC, and audit trails.
- **2.2 Content Catalog & Creator Management:** Manages content assets, metadata, creator profiles, and publication status.

- **2.3 Subscription & Plan Management:** Handles subscription products, upgrades, renewals, and access entitlements.
- **2.4 Editorial & Content Scheduling:** Manages content curation, review workflows, and publication calendars.
- **2.5 Content Licensing & Rights Management:** Tracks content licenses, territory restrictions, and expiry management.
- **2.6 Royalty & Revenue Management:** Calculates creator royalties, manages payout schedules, and tracks revenue sharing.
- **2.7 Media Analytics & Reporting:** Provides dashboards for audience reach, content performance, and revenue metrics.
- **2.8 Notifications & Alerts:** New content alerts, subscription renewal reminders, and royalty payout notifications.

3. Architecture Overview

- **Frontend:** Angular or React
- **Backend:** REST API-based architecture
- **Database:** Relational Database (MySQL / PostgreSQL / SQL Server)

4. Module-Wise Design

4.1 Identity & Access Management Module

Features

- Subscriber, creator, and staff registration, login, and role-based access control
- Tiered content access enforcement based on subscription level and territory

Entities

- **User**
 - UserID
 - Name
 - Role (Subscriber/Creator/Editorial/RightsManager/RevenueAnalyst/Admin)
 - Email
 - Phone
 - Country

- Status (Active/Suspended/Inactive)
- **AuditLog**
 - AuditID
 - UserID
 - Action
 - EntityType
 - Timestamp

4.2 Content Catalog & Creator Management Module

Features

- Onboard creators and manage their content catalog with rich metadata
- Track content versions, format variants, and publication history

Entities

- **Creator**
 - CreatorID
 - UserID
 - DisplayName
 - Genre
 - Country
 - RoyaltyTier
 - BankAccountRef
 - Status (Active/Suspended/PendingReview)
- **ContentAsset**
 - ContentID
 - CreatorID
 - Title
 - Type (Video/Audio/Article/Podcast/Ebook)
 - Genre

- Language
- DurationSeconds
- Synopsis
- FilePath
- ThumbnailPath
- Status (Draft/UnderReview/Published/Archived/Removed)
- **ContentTag**
 - TagID
 - ContentID
 - TagName
 - TagCategory (Genre/Mood/Theme/AgeRating)

4.3 Subscription & Plan Management Module

Features

- Create tiered subscription plans with content access levels and pricing
- Handle subscriptions, renewals, upgrades, downgrades, and cancellations

Entities

- **SubscriptionPlan**
 - PlanID
 - Name (Free/Basic/Standard/Premium)
 - Price
 - BillingCycle (Monthly/Quarterly/Annual)
 - ContentAccessLevel
 - MaxDevices
 - DownloadAllowed
 - Status (Active/Discontinued)
- **UserSubscription**
 - SubscriptionID

- UserID
- PlanID
- StartDate
- EndDate
- RenewalType (AutoRenew/Manual)
- Status (Active/Expired/Cancelled/Suspended)

4.4 Editorial & Content Scheduling Module

Features

- Review and approve creator-submitted content before publication
- Schedule featured placements and curate content collections

Entities

- **EditorialReview**
 - ReviewID
 - ContentID
 - ReviewerID
 - SubmissionDate
 - ReviewDate
 - Decision (Approved/Rejected/RevisionRequired)
 - Remarks
 - Status (Pending/Completed)
- **ContentCollection**
 - CollectionID
 - Name
 - Category (Featured/Trending/Curated/New)
 - ContentIDs
 - PublishDate
 - ExpiryDate

- Status (Scheduled/Active/Expired)
- **PublicationSchedule**
 - ScheduleID
 - ContentID
 - PublishDateTime
 - ExpiryDateTime
 - Territory
 - Status (Scheduled/Published/Expired/Cancelled)

4.5 Content Licensing & Rights Management Module

Features

- Record content licensing agreements and territory restriction rules
- Track license expiry dates and manage content takedowns on expiry

Entities

- **LicenseAgreement**
 - LicenseID
 - ContentID
 - LicensorID
 - LicenseeRef
 - Territory
 - RightsType (Exclusive/Non-Exclusive/Limited)
 - StartDate
 - EndDate
 - LicenseFee
 - Status (Active/Expired/Terminated)
- **TerritoryRestriction**
 - RestrictionID
 - ContentID

- RestrictedCountries
- AllowedCountries
- EffectiveDate
- Status (Active/Inactive)

4.6 Royalty & Revenue Management Module

Features

- Calculate creator royalties based on views, streams, and configured revenue-share rules
- Generate payout schedules and track disbursement status

Entities

- **RoyaltyRule**
 - RuleID
 - CreatorTier
 - RevenueSharePercent
 - MinimumPayoutThreshold
 - PayoutFrequency (Monthly/Quarterly)
 - EffectiveDate
 - Status (Active/Inactive)
- **RoyaltyStatement**
 - StatementID
 - CreatorID
 - Period
 - TotalViews
 - TotalRevenue
 - RoyaltyAmount
 - Status (Draft/Finalised/Paid)
- **RoyaltyPayout**
 - PayoutID

- StatementID
- CreatorID
- Amount
- PayoutDate
- Method (BankTransfer/WalletCredit)
- Status (Pending/Processed/Failed)

4.7 Media Analytics & Reporting Module

Features

- Audience reach, content views, and watch-time dashboards by genre, language, and period
- Subscription revenue breakdown, creator performance rankings, and royalty payout summaries

Entities

- **MediaReport**
 - ReportID
 - Scope (Creator/Genre/Plan/Period/Territory)
 - Metrics (TotalViews, UniqueViewers, AvgWatchTime, SubscriptionRevenue, RoyaltyPaid, ChurnRate)
 - GeneratedDate

4.8 Notifications & Alerts Module

Features

- New content release alerts and subscription renewal reminders for subscribers
- Editorial review outcomes, royalty statement availability, and license expiry alerts for creators and rights managers

Entities

- **Notification**
 - NotificationID
 - UserID

- Message
- Category (Content/Subscription/Royalty/License/Editorial)
- Status (Unread/Read/Dismissed)
- CreatedDate

5. Deployment Strategy

- **Local:** Angular/React frontend, Spring Boot/.NET Core backend, local DB
- **Production:** Cloud or on-premises with secure APIs and stateless services

6. Database Design

Tables:

- **User**(UserID, Name, Role, Email, Phone, Country, Status)
- **AuditLog**(AuditID, UserID, Action, EntityType, Timestamp)
- **Creator**(CreatorID, UserID, DisplayName, Genre, Country, RoyaltyTier, BankAccountRef, Status)
- **ContentAsset**(ContentID, CreatorID, Title, Type, Genre, Language, DurationSeconds, Synopsis, FilePath, ThumbnailPath, Status)
- **ContentTag**(TagID, ContentID, TagName, TagCategory)
- **SubscriptionPlan**(PlanID, Name, Price, BillingCycle, ContentAccessLevel, MaxDevices, DownloadAllowed, Status)
- **UserSubscription**(SubscriptionID, UserID, PlanID, StartDate, EndDate, RenewalType, Status)
- **EditorialReview**(ReviewID, ContentID, ReviewerID, SubmissionDate, ReviewDate, Decision, Remarks, Status)
- **ContentCollection**(CollectionID, Name, Category, ContentIDs, PublishDate, ExpiryDate, Status)
- **PublicationSchedule**(ScheduleID, ContentID, PublishDateTime, ExpiryDateTime, Territory, Status)
- **LicenseAgreement**(LicenseID, ContentID, LicenserID, LicenseeRef, Territory, RightsType, StartDate, EndDate, LicenseFee, Status)
- **TerritoryRestriction**(RestrictionID, ContentID, RestrictedCountries, AllowedCountries, EffectiveDate, Status)

- **RoyaltyRule**(RuleID, CreatorTier, RevenueSharePercent, MinimumPayoutThreshold, PayoutFrequency, EffectiveDate, Status)
- **RoyaltyStatement**(StatementID, CreatorID, Period, TotalViews, TotalRevenue, RoyaltyAmount, Status)
- **RoyaltyPayout**(PayoutID, StatementID, CreatorID, Amount, PayoutDate, Method, Status)
- **MediaReport**(ReportID, Scope, Metrics, GeneratedDate)
- **Notification**(NotificationID, UserID, Message, Category, Status, CreatedDate)

7. User Interface Design

- **Subscriber Portal:** Content catalog browser, subscription manager, watchlist, viewing history
- **Creator Studio:** Content upload form, catalog manager, performance metrics dashboard, royalty statement viewer
- **Editorial Console:** Review queue, content approval workflow, collection builder, publication calendar
- **Rights Manager Dashboard:** License registry, territory restriction editor, expiry alert tracker
- **Revenue Dashboard:** Subscription revenue breakdown, royalty calculation run, payout tracker
- **Admin Console:** Plan configurator, content policy rules, creator tier management, report scheduler

8. Non-Functional Requirements

- **Performance:** Support 50,000+ concurrent subscriber users during new season or content release peaks
- **Security:** Content DRM reference enforcement, subscriber PII protection, RBAC, and full audit trails
- **Scalability:** Multi-territory deployment with region-specific content catalogs and pricing
- **Availability:** 99.9% uptime with content catalog reads served from cached data for resilience
- **Maintainability:** Configurable subscription tiers, royalty rules, and editorial workflows without code changes

- **Observability:** Real-time content performance dashboards, subscription churn alerts, and royalty pipeline metrics

9. Assumptions & Constraints

- Phase 1 excludes actual video/audio streaming delivery; content is referenced via internal file paths
- Payment processing for subscriptions is simulated internally; payment gateway integration is deferred
- All notifications are in-app only; email digest and push notification integrations are deferred to Phase 2
- Viewership data is simulated via manual entry; CDN-level analytics feed integration is out of Phase 1 scope
- Fully implementable using standard Java Spring Boot / ASP.NET Core, Angular/React, and RDBMS tooling